

Design & development of robotics arm

Problem statement: A 3-DOF robotic arm is required to achieve accurate (± 10 mm) and controlled positioning of its end-effector in three-dimensional space ensuring smooth and repeatable motion (± 5 mm) within its mechanical limits (Max. reach 370 mm & min. reach 20mm).

List of Targets

Torque sizing margin: Each servo motor should provide at least the maximum torque required to move the arm through its full range of motion without stalling, ensuring reliable and safe operation. (> 1.5 N-m)

Positioning Accuracy: The end-effector must be positioned within ± 10 mm of the commanded position.

Repeatability: The arm should return to the same position consistently when the same command is executed minimum 5 times (< 5 mm)

Homing Repeatability: The ability of the robotic arm to return to the same predefined home position each time the homing procedure is performed.

Payload Capacity: The robotic arm is intended only for controlled positioning of its end-effector and is not designed to carry external loads. (Maximum 300 gm).

Runtime: The duration for which the robotic arm can operate continuously under normal operating conditions without overheating, excessive performance degradation, or requiring a restart. (Minimum 30 minutes of continuous operation during a test session.)

List of constraint

Workspace Constraint: The robotic arm must operate within the specified workspace.

- Maximum Reach: 370 mm
- Minimum Reach: 20 mm

Joint Angle Constraint:

Each joint must operate within its specified angular limits to prevent mechanical interference and damage.

- Joint 1 – Base Rotation: 0° – 360°
- Joint 2 – Shoulder: 0° – 180°
- Joint 3 – Elbow: 0° – 180°

Power Constraint:

The robotic arm must operate using the specified power supply, and the motors and controller must remain within their rated voltage and current limits. (12V)

Safety Constraint:

The arm must operate within defined joint limits and should stop or prevent movement when a mechanical or electrical limit is reached. The system should also avoid excessive motor load and overheating.

Identify User and Context**User**

An engineering student or laboratory operator who operates the 3-DOF robotic arm through a computer interface to command the end-effector to predefined positions.

Context

The robotic arm is bench-mounted and operated indoors in a laboratory environment for testing, experimentation, and demonstration of controlled positioning and joint movement. It is intended for educational and experimental use, not for industrial production.

How it will be measured

Torque Sizing Margin: The required torque for each servo motor will be calculated based on the arm-link length, mass of the links, payload, and gravitational force. The calculated maximum required torque will then be compared with the servo motor's rated torque. The selected servo motor should provide a torque greater than the calculated requirement, with the target value being $> 1.5 \text{ N}\cdot\text{m}$.

Positioning Accuracy: The positioning accuracy will be measured by commanding the end-effector to move to predefined target positions. The actual position of the end-effector will be measured using a ruler, measuring scale, or coordinate reference system. The difference between the commanded position and actual position will be calculated. The maximum position error should remain within $\pm 10 \text{ mm}$.

Repeatability: Repeatability will be measured by commanding the robotic arm to the same predefined position a minimum of 5 times. The actual end-effector position will be recorded after each movement. The positions obtained from all trials will be compared, and the maximum deviation between the repeated positions will be calculated. The arm will meet the target if the deviation is less than 5 mm.

Homing Repeatability: Homing repeatability will be measured by moving the robotic arm away from its predefined home position and then performing the homing operation. The final position of each joint after homing will be recorded. This process will be repeated a minimum

of 5 times, and the variation in the final home position will be compared. The arm should return consistently to the same predefined home position within the specified tolerance.

Payload Capacity: Payload capacity will be measured by attaching a known load to the end-effector and operating the arm through its required range of motion. The load will be increased up to the maximum specified value of 300 g. The payload requirement will be considered satisfied if the robotic arm can move and position the end-effector without servo stalling, excessive vibration, loss of position, or mechanical deformation.

Runtime: Runtime will be measured by operating the robotic arm continuously through a predefined sequence of movements for a minimum of 30 minutes. The start and end times will be recorded, while the arm's movement, motor temperature, and controller performance will be monitored during the test. The requirement will be satisfied if the robotic arm completes the 30-minute operation without overheating, stalling, requiring a restart, or showing significant performance degradation.

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